

Week 2 – §3.1 & 3.2: Matrices en Gauss–Jordan

Studiegids uit de HC-aantekeningen (dictaat-nummers ter referentie)

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Lecture framing (slides 1–9): “Waar gaat lineaire algebra over?” Answer for now: matrices, and using them to solve systems of linear equations via the Gauss–Jordan algorithm.

Convention for the rest of the course: $i, j, k, n, m \in \mathbb{N}$ with value ≥ 1 .

1. What is a matrix? (Def. 3.1.1)

An $m \times n$ **matrix** A consists of $m \cdot n$ numbers from $\mathbb{R}$ (or $\mathbb{C}$) arranged in m **rijen** (rows) and n **kolommen** (columns):

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}, \quad a_{ij} \in \mathbb{R} : i\text{-th row, } j\text{-th column.}$$

Also written A_{ij} . Mnemonic: **rows first, then columns** — “ $m \times n$ ” = rows $\times$ columns.

The set of all $m \times n$ matrices is $M_{m,n}$ (dictaat: M_{mn}).

Merk op: $M_{m,1} = \mathbb{R}^m$ — a column vector *is* an $m \times 1$ matrix: $\{(a_{11}, \dots, a_{m1})^t \mid a_{i1} \in \mathbb{R}\}$.

2. Bewerkingen met matrices (§3.1)

2.1 Optelling (Def. 3.1.3)

Only defined for $A, B \in M_{m,n}$ with the **same format**. It works **componentsgewijs**:

$$(A + B)_{ij} = A_{ij} + B_{ij}.$$

VB:

$$\begin{pmatrix} -1 & 0 & 3 \\ 6 & 2 & 11 \end{pmatrix} + \begin{pmatrix} 0 & 1 & 1 \\ 3 & 0 & 1 \end{pmatrix} = \begin{pmatrix} -1 & 1 & 4 \\ 9 & 2 & 12 \end{pmatrix}.$$

2.2 Scalaire vermenigvuldiging (Def. 3.1.3)

Also componentsgewijs. For $\lambda \in \mathbb{R}$, $A \in M_{m,n}$:

$$(\lambda A)_{ij} = \lambda A_{ij}.$$

VB:

$$3 \begin{pmatrix} 0 & -4 \\ 1 & 1 \\ 2 & 8 \end{pmatrix} = \begin{pmatrix} 0 & -12 \\ 3 & 3 \\ 6 & 24 \end{pmatrix}.$$

(The rules for $+$ and $\lambda \cdot$ are the same as for vectors: Stelling 3.1.4.)

2.3 Vermenigvuldiging (Def. 3.1.5) — the one that needs care

AB is **only defined** when

$$\# \text{kolommen van } A = \# \text{rijen van } B.$$

If A is $m \times n$ and B is $n \times k$, then AB is $m \times k$. The “inner” n ’s must match and disappear; the “outer” sizes survive:

$$\begin{array}{ccccc} A & & B & & AB \\ m \times n & n \times k & \rightarrow & m \times k \\ & \wedge \text{ must match} & & & \end{array}$$

Formula: $(AB)_{ij} = (i\text{-th row of } A) \cdot (j\text{-th column of } B)$, an inner product:

$$(AB)_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj} = \sum_{k=1}^n a_{ik}b_{kj}.$$

VB (from the board):

$$\begin{pmatrix} 1 & 0 & 3 \\ -1 & 2 & 5 \end{pmatrix} \begin{pmatrix} 1 & 0 & 2 & 1 \\ -1 & 1 & 0 & 1 \\ 3 & 0 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 10 & 0 & 5 & 4 \\ 12 & 2 & 3 & 6 \end{pmatrix}.$$

E.g. entry $(1,1)$: row $(1,0,3) \cdot$ column $(1,-1,3) = 1 - 0 + 9 = 10$. Entry $(2,4)$: $(-1,2,5) \cdot (1,1,1) = -1 + 2 + 5 = 6$. A 2×3 times a 3×4 gives a 2×4 .

Slide Q1. $A = \begin{pmatrix} 1 & 0 \\ 0 & 4 \\ 2 & -1 \end{pmatrix}$, $B = \begin{pmatrix} 3 & -1 & 2 & 0 \\ 0 & 1 & 2 & 3 \end{pmatrix}$. A is 3×2 , B is 2×4 , so AB is 3×4 :

$$AB = \begin{pmatrix} 3 & -1 & 2 & 0 \\ 0 & 4 & 8 & 12 \\ 6 & -3 & 2 & -3 \end{pmatrix}, \quad (AB)_{2,3} = (0,4) \cdot (2,2) = 8.$$

(BA would be 2×4 times 3×2 : not defined.)

2.4 Matrix multiplication is NOT commutative (slide Q2)

- If A is $n \times m$ and B is $m \times n$ with $n \neq m$: AB is $n \times n$ but BA is $m \times m$, so $AB \neq BA$ trivially (different sizes).
- Even for square matrices it usually fails. Lecture example (“blauw”):

$$\begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix},$$

so $AB \neq BA$.

- Sometimes it *does* hold (“zwart”/“geel”): $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ — the identity matrix commutes with everything (Def. 3.1.6: such A, B *commuteren*).

Conclusie: matrixvermenigvuldiging is niet commutatief. Never swap factors. (What *does* hold: associativity $A(BC) = (AB)C$, distributivity, $IA = AI = A$ — Stelling 3.1.7.)

3. Gauss–Jordan eliminatie (§3.2)

Goal. Find *all* $x_1, x_2, x_3 \in \mathbb{R}$ with

$$\begin{aligned} x_1 + x_2 - x_3 &= 1 \\ 2x_1 + 4x_2 - x_3 &= 4 \\ -x_1 + 5x_2 + 6x_3 &= 9 \end{aligned}$$

3.1 Which operations keep the solution set unchanged? (Stelling 3.2.5)

1. **Vergelijkingen verwisselen** (swap two equations).
2. **Een vergelijking links en rechts vermenigvuldigen met een getal $\neq 0$.**
3. **Een veelvoud van een vergelijking optellen bij een andere vergelijking.**

Two systems related by these moves are **equivalent** (notation $\sim$): they have *exactly* the same oplossingsverzameling.

3.2 The algorithm: Gauss-eliminatie

- a) Put at the top an equation whose coefficient of x_1 is $\neq 0$.
- b) Use it to “wegvegen” (sweep away, make coefficient 0) every x_1 in the equations below.
- c) Do the same for $x_2, x_3, \dots$ in rows 2, 3, $\dots$

Applied to the example:

$$\begin{array}{lll} x_1 + x_2 - x_3 = 1 & & x_1 + x_2 - x_3 = 1 \\ 2x_2 + x_3 = 2 & (\text{r2} - 2\text{r1}) & \sim \quad 2x_2 + x_3 = 2 \\ 6x_2 + 5x_3 = 10 & (\text{r3} + \text{r1}) & \quad 2x_3 = 4 \quad (\text{r3} - 3\text{r2}) \end{array} \quad (*)$$

Up to here it is called **Gauss-eliminatie**: you cannot sweep further *downwards*. Now two options:

Methode 1 — terugsubstitutie (from bottom to top): $2x_3 = 4 \Rightarrow x_3 = 2$; $2x_2 + 2 = 2 \Rightarrow x_2 = 0$; $x_1 + 0 - 2 = 1 \Rightarrow x_1 = 3$.

Methode 2 — Jordan-eliminatie: Gauss in the *opposite* direction (start at the bottom row and last variable, sweep *upwards*):

$$\begin{array}{rcl}
 x_1 + x_2 - x_3 = 1 & x_1 + x_2 = 3 & x_1 = 3 \\
 (*) \sim 2x_2 + x_3 = 2 & \sim 2x_2 = 0 & \sim x_2 = 0 \\
 x_3 = 2 & x_3 = 2 & x_3 = 2
 \end{array}$$

Either way: $(x_1, x_2, x_3)^t = (3, 0, 2)^t$.

3.3 In matrix language

The system is $A\mathbf{x} = \mathbf{b}$:

$$\begin{pmatrix} 1 & 1 & -1 \\ 2 & 4 & -1 \\ -1 & 5 & 6 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 4 \\ 9 \end{pmatrix}.$$

Shorthand: the **uitgebreide (extended) coëfficiëntenmatrix** $(A \mid \mathbf{b})$:

$$\left(\begin{array}{ccc|c} 1 & 1 & -1 & 1 \\ 2 & 4 & -1 & 4 \\ -1 & 5 & 6 & 9 \end{array} \right) \quad \begin{array}{l} \text{left block: de coëfficiëntenmatrix } A \\ \text{right column: de uitkomstvector } \mathbf{b} \end{array}$$

Row operations $r_i \rightarrow \dots$ are the three allowed moves. Gauss–Jordan becomes:

$$\begin{aligned}
 & \left(\begin{array}{ccc|c} 1 & 1 & -1 & 1 \\ 2 & 4 & -1 & 4 \\ -1 & 5 & 6 & 9 \end{array} \right) \xrightarrow[r_3 \rightarrow r_3 + r_1]{r_2 \rightarrow r_2 - 2r_1} \left(\begin{array}{ccc|c} 1 & 1 & -1 & 1 \\ 0 & 2 & 1 & 2 \\ 0 & 6 & 5 & 10 \end{array} \right) \\
 & \xrightarrow{r_3 \rightarrow r_3 - 3r_2} \left(\begin{array}{ccc|c} 1 & 1 & -1 & 1 \\ 0 & 2 & 1 & 2 \\ 0 & 0 & 2 & 4 \end{array} \right) \xrightarrow{r_3 \rightarrow r_3/2} \left(\begin{array}{ccc|c} 1 & 1 & -1 & 1 \\ 0 & 2 & 1 & 2 \\ 0 & 0 & 1 & 2 \end{array} \right)
 \end{aligned}$$

Tot zover Gauss-eliminatie. Then Jordan:

$$\xrightarrow[r_2 \rightarrow r_2 - r_3]{r_1 \rightarrow r_1 + r_3} \left(\begin{array}{ccc|c} 1 & 1 & 0 & 3 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 2 \end{array} \right) \xrightarrow{r_2 \rightarrow r_2/2} \left(\begin{array}{ccc|c} 1 & 1 & 0 & 3 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 2 \end{array} \right) \xrightarrow{r_1 \rightarrow r_1 - r_2} \left(\begin{array}{ccc|c} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 2 \end{array} \right)$$

Read off: $1x_1 + 0x_2 + 0x_3 = 3$, $0x_1 + 1x_2 + 0x_3 = 0$, $0x_1 + 0x_2 + 1x_3 = 2$, i.e. $\mathbf{x} = (3, 0, 2)^t$.

3.4 Terminologie (Def. 3.2.6, 3.2.8)

- After Gauss-eliminatie the matrix is in **rijgereduceerde vorm** (trapvorm). Toelichting: for every row (except the first) either
 - the number of **leidende nullen** (leading zeros) is strictly larger than in the row above,
 - or**
 - the row consists of zeros only.
- The first element $\neq 0$ in a row is a **pivot** (pivotelement).
- A variable x_j is a **pivotvariabele** if column j contains a pivot. In the example x_1, x_2, x_3 are all pivot variables.
- After Jordan as well (pivots = 1, zeros above *and* below each pivot) the matrix is **volledig rijgereduceerd**.

*Gauss-eliminatie alone is already enough to analyse a system (how many solutions?).
Jordan just makes reading off the solution trivial.*

4. The role of pivots: three examples

VB1 — all variables are pivot variables → unique solution

After Gauss–Jordan: $\left(\begin{array}{ccc|c} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 2 \end{array}\right)$. Pivots in columns $x_1, x_2, x_3 \rightarrow$ every value is fixed: $(3, 0, 2)^t$.

VB2 — a row of zeros: strijdig or a free variable

$$\begin{array}{l} x_1 - 2x_2 = 1 \\ -2x_1 + 4x_2 = a \end{array} \quad (a \in \mathbb{R}), \quad \left(\begin{array}{cc|c} 1 & -2 & 1 \\ -2 & 4 & a \end{array}\right) \xrightarrow{r_2 \rightarrow r_2 + 2r_1} \left(\begin{array}{cc|c} 1 & -2 & 1 \\ 0 & 0 & a+2 \end{array}\right)$$

Pivot only in column 1: x_1 is a pivot variable, x_2 is **not**. The last row says $0x_1 + 0x_2 = a + 2$.

Gevalsonderscheiding:

- $a \neq -2$: $0 = a + 2 \neq 0$ is impossible $\rightarrow$ the system is **strijdig** (Def. 3.2.3: no solution).
General rule: a row of all zeros on the left with a non-zero number on the right $\Rightarrow$ strijdig.
- $a = -2$: the row reads $0 = 0$ and gives no information. The non-pivot variable x_2 can be chosen freely: put $x_2 = t$, $t \in \mathbb{R}$. Then $x_1 - 2t = 1$, so $x_1 = 1 + 2t$. Solution set:

$$\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 1 + 2t \\ t \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} + t \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \quad t \in \mathbb{R} \quad \Rightarrow \quad \text{a line in the plane.}$$

Opmerkingen (from the board):

1. A zero row: strijdig if the right-hand side $\neq 0$; “no information” if it is 0.
2. **Niet-pivotvariabelen zijn vrij te kiezen. Pivotvariabelen liggen vast** once the non-pivot variables have been chosen.

VB3 — several free variables (5 unknowns, 3 equations)

$$\begin{array}{l} x_1 + 2x_2 + x_4 + x_5 = 1 \\ 2x_1 + 4x_2 + x_3 + 4x_4 + 3x_5 = 1 \\ -x_1 - 2x_2 + x_3 + x_4 = -2 \end{array}$$

$$\begin{aligned} &\left(\begin{array}{ccccc|c} 1 & 2 & 0 & 1 & 1 & 1 \\ 2 & 4 & 1 & 4 & 3 & 1 \\ -1 & -2 & 1 & 1 & 0 & -2 \end{array}\right) \xrightarrow[r_3 \rightarrow r_3 + r_1]{r_2 \rightarrow r_2 - 2r_1} \left(\begin{array}{ccccc|c} 1 & 2 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 2 & 1 & -1 \\ 0 & 0 & 1 & 2 & 1 & -1 \end{array}\right) \\ &\xrightarrow{r_3 \rightarrow r_3 - r_2} \left(\begin{array}{ccccc|c} \boxed{1} & 2 & 0 & 1 & 1 & 1 \\ 0 & 0 & \boxed{1} & 2 & 1 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right) \end{aligned}$$

This is already volledig rijgereduceerd. Pivots in columns 1 and 3, so:

- **Niet strijdig** (zero row has 0 on the right).
- x_1, x_3 are pivot variables $\rightarrow$ fixed once the others are chosen.
- x_2, x_4, x_5 are not pivot variables $\rightarrow$ **vrij te kiezen**: $x_2 = s$, $x_4 = t$, $x_5 = u$.

Back-substitute: $x_3 + 2t + u = -1 \Rightarrow x_3 = -1 - 2t - u$; $x_1 + 2s + t + u = 1 \Rightarrow x_1 = 1 - 2s - t - u$.
Oplossingsverzameling:

$$\left\{ \begin{pmatrix} 1 - 2s - t - u \\ s \\ -1 - 2t - u \\ t \\ u \end{pmatrix} \middle| s, t, u \in \mathbb{R} \right\} = \left\{ \begin{pmatrix} 1 \\ 0 \\ -1 \\ 0 \\ 0 \end{pmatrix} + s \begin{pmatrix} -2 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} + t \begin{pmatrix} -1 \\ 0 \\ -2 \\ 1 \\ 0 \end{pmatrix} + u \begin{pmatrix} -1 \\ 0 \\ -1 \\ 0 \\ 1 \end{pmatrix} \middle| s, t, u \in \mathbb{R} \right\}$$

→ a **3-dimensional “vlak” of solutions in $\mathbb{R}^5$** : one steunvector plus three direction vectors, exactly the week-1 pattern $\mathbf{p} + \lambda \mathbf{a} + \mu \mathbf{b} + \dots$

*Pattern to memorise: **#free parameters = #variables – #pivots**. Each free variable contributes one direction vector to the solution set.*

Decision table after Gauss elimination

What you see in $(A' \mid \mathbf{b}')$	Conclusion
A row $(0 \ \dots \ 0 \mid c)$ with $c \neq 0$	Strijdig : no solution
No such row, every variable is a pivot variable	Exactly one solution
No such row, some variable is not a pivot variable	Infinitely many solutions; free variables = parameters

Zelftest

1. $A \in M_{3,2}$, $B \in M_{2,5}$. Is AB defined? BA ? Sizes? *Answer*: AB yes, 3×5 . BA : 2×5 times 3×2 , $5 \neq 3$, not defined.
2. Compute $(AB)_{3,1}$ for the Q1 matrices. *Answer*: row 3 of A is $(2, -1)$, column 1 of B is $(3, 0)$: 6.
3. Give the three row operations and say why each keeps the solution set. *Answer*: swap rows (same equations, other order); multiply a row by $c \neq 0$ (divide by c to get back); add a multiple of one row to another (subtract it again to get back). Each is reversible, so no solutions are gained or lost.
4. Reduce $\left(\begin{array}{cc|c} 2 & 4 & 2 \\ 1 & 2 & 3 \end{array} \right)$ and classify. *Answer*: $r_1 \rightarrow r_1/2$ gives $(1 \ 2 \mid 1)$; $r_2 \rightarrow r_2 - r_1$ gives $(0 \ 0 \mid 2)$. Row $0 = 2 \Rightarrow$ strijdig.
5. In VB3, why are there exactly three direction vectors? *Answer*: 5 variables, 2 pivots, so $5 - 2 = 3$ free variables s, t, u .